

Rituraj Kulshresth

GitHub | Portfolio

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SUMMARY

Software engineer specializing in enterprise AI platforms. Designed and shipped three production agent and RAG platforms at Warner Bros. Discovery used by 11+ internal teams. Architecture lead on organization-wide documents for agent authentication, evaluation, and retrieval.

EDUCATION

- **Indian Institute of Technology, Jodhpur** Jodhpur, Rajasthan
Bachelor of Technology in Computer Science and Engineering; July 2018 – June 2022

EXPERIENCE

- **Warner Bros. Discovery** Hyderabad, India
Software Development Engineer 2 (AI Platforms) Nov 2024 - Present
- **ACME (Agent Core for Managed Execution): Enterprise Agent Platform:**
 - Designed the credentials and authentication architecture for ACME: a **four-resource model** (Team, App, Agent, Sub-agent) with separate Edit and Invocation ACLs, dual auth modes (**act_as_app** for production via API keys and AWS Secrets Manager; **act_as_user** for the developer playground via Okta OIDC), and a **sub-agent safety property** where only the caller's identity flows through every tool call, making cross-team agent reuse safe by construction.
 - Built the **ACME Playground**, a sandboxed surface that lets developers prompt-test agents, inspect tool calls and streaming events, and validate sub-agent orchestration before publishing the agent for reuse by apps organization-wide.
 - Refactored ACME agent configuration off local YAML files into a **PostgreSQL (JSONB) registry** with versioning, a draft/published/archived lifecycle, and an API-driven Console Builder and Catalog, enabling **multi-tenant onboarding** without code changes.
 - Shipped **multi-model serving** via AWS Bedrock inference profiles alongside OpenAI and Anthropic providers in ACME, standardizing model routing and fallback behaviour for production traffic.
 - Drove ACME from **CRITICAL**-severity SAST findings to **zero** in a single sprint, hardening the platform's release posture across all environments ahead of broader org rollout.
 - Provisioned ACME infrastructure on the org's IaC platform (Terraform/Terragrunt, Helm) and bootstrapped the **BOLT dual-language monorepo** (FastAPI runtime + Next.js console) with CI/CD across DEV, INT, STG, and PROD.
- **Knowledge Hub: Enterprise RAG-as-a-Service:**
 - Designed and built Knowledge Hub **end-to-end: a self-serve retrieval service** built on **AWS OpenSearch** where teams create their own enterprise document indices and share both the document catalog and the index catalog across teams, so any team building an AI app can reuse existing indices without re-ingesting. Now consumed by multiple internal apps; authored the org-wide architecture document for the same.
 - Implemented ingestion from Confluence, Jira, GitHub, S3, Airflow logs, and arbitrary text sources, with bulk-upload support, team-specific chunking strategies, and a **semantic chunking** mode that splits content by meaning instead of token windows.
 - Added **multimodal embedding support**: AWS Bedrock Titan and Anthropic Claude embeddings for text/code, and a Bedrock multimodal model for image ingestion, broadening access to teams whose documents live as diagrams, screenshots, and scanned PDFs.
- **Daisy: Data Analytics AI Platform:**
 - Developed Daisy, an internal **multi-agent AI platform** used by **11+** engineering and business teams, to build domain-specific chatbots with fine-grained control over tools, data sources (SQL, Text), and model configurations using Vercel AI SDK, FastAPI and LangChain.
 - Cut Daisy's monthly infra spend from **~\$700 to ~\$200 per environment** across DEV/INT/STG and from **~\$1000 to ~\$800 in PROD** by analysing traffic and workload patterns, right-sizing compute/storage, and decommissioning unused services.
 - Migrated all Daisy AWS Lambdas off the legacy Discovery AWS account onto the org's standardised IaC AWS accounts with **zero downtime** during the cutover.
 - Built an evaluation framework using DeepEval and custom metrics to measure prompt quality and LLM correctness, improving evaluation accuracy by **27%**.
 - Achieved **~66%** reduction in storage size by optimizing how chat sessions were logged with Vercel AI SDK, eliminated redundancy and improved OpenSearch indexing and query performance.
 - Led the migration of the entire platform to a new in-house IaC environment during an org split: migrated codebases, provisioned infrastructure using Terraform/Terragrunt, and maintained **zero downtime** across all environments.

- Authored multiple **Architecture Documents** (Evaluation Platform, Daisy Vector DB API) published **organization-wide**, and conducted code reviews for a team of **7 engineers**.

- **Deloitte**

Hyderabad, India

July 2022 – Nov 2024

- **Outlook add-ins and Document Archival Apps:**

- Developed and deployed S2SACC Outlook add-in using Yeoman and MS GraphAPI, facilitating document transfer to ContentCloud via email, with role-based access based on distribution groups, serving **1000+ users**.
- Developed S2EDS Outlook add-in used to send documents to enterprise servers from mailbox using ReactJs and Yeoman, serving **500+ users**.
- Migrated services from **Mule ESB to AWS Lambda**, reducing compute costs and improving performance for the Document Archival system. Maintained **4 critical Java applications**, ensuring seamless business operations.

PROJECTS

- **RustyBun:** Published RustyBun, a clipboard management application designed to streamline your workflow by providing convenient access to clipboard history on Debian-based Linux distributions
- **X86 Architecture OS:** Built a 32-bit operating system with keyboard input, VGG output driver, and a functional shell for basic tasks using Assembly and C
- **RGB face video data to Core body Temperature:** Implemented CNN models using TensorFlow for core body temperature prediction using diverse image data (NIR, Hyper-spectral, RGB) to achieve **60%** accuracy

SKILLS

- **Languages:** Python, C/C++, JavaScript/TypeScript, Rust, Java, Assembly, SQL
- **AI & Agent Frameworks:** LangGraph, LangChain, LangSmith, OpenAI SDK, Anthropic Claude SDK, Vercel AI SDK, DeepEval, AWS Bedrock (Titan, Claude, multimodal embeddings), TensorFlow
- **Backend & Frontend:** FastAPI, SQLAlchemy, Alembic, Next.js, React, PostgreSQL, MERN, Electron, Yeoman
- **Cloud & DevOps:** AWS (EKS, OpenSearch, Lambda, S3, API Gateway, Secrets Manager), Kubernetes, Helm, Docker, Terraform, Terragrunt, Okta OIDC, Git, Mainframe (z/OS)

ACCOLADES

- **KVPY:** Recognized as a Kishore Vaigyanik Protsahan Yojana Scholar in 2017-2018
- **Deloitte:** Honored with 3 Excellence Awards for development of the S2SACC Add-in and the OpenText Server upgrade
- **Warner Bros. Discovery:** Honored with an Excellence Award for designing and building Knowledge Hub